A CHANNEL MIX METHOD FOR FINE-GRAINED CROSS-MODAL RETRIEVAL

Yang Shen[1,2], *Xuhao Sun*[1], *Xiu-Shen Wei*[1,2,3,*], *Hanxu Hu*[4], *Zhipeng Chen*[5]

[1]School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China
[2]State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China
[3]State Key Laboratory for Novel Software Technology, Nanjing University, Nanjing, China
[4]The University of Edinburgh, Edinburgh, United Kingdom
[5]National Internet Emergency Center (CNCERT/CC), Beijing, China

ABSTRACT

In this paper, we propose a simple but effective method for dealing with the challenging fine-grained cross-modal retrieval task where it aims to enable flexible retrieval among subordinate categories across different modalities. Specifically, in order to enhance information interaction in different modalities for fine-grained objects, a channel mix method is developed and performed upon the channels of deep activations across different modalities. After that, a 1×1 convolution is employed to aggregate the mixed channels into a unified feature vector. Moreover, equipped with a novel fine-grained cross-modal center loss, our method can further improve the intra-class separability as well as inter-class compactness for multi-modalities. Experiments are conducted on the fine-grained cross-modal benchmark dataset and show our superiority over competing methods. Meanwhile, ablation studies also demonstrate the effectiveness of our proposals.

Index Terms— Fine-Grained Retrieval, Cross-Modal Retrieval, Cross-Modal Channel Mix (CMCM).

1. INTRODUCTION

With the development of relevant technologies, videos, images, texts and audios have become the main form of humans understanding the world and communicating with each other. The rapid growth of multi-modal data has brought about a large number of application requirements for cross-modal retrieval, where the goal is to enable interactivity across content modalities. Compared with single-modal retrieval, cross-modal retrieval presents a higher technical challenge because of the large inter-modal variations [1, 2, 3]. Thus, more theories and techniques for cross-modal retrieval are required to be developed.

Existing researches on cross-modal retrieval mainly focus on the image-and-text modality [4, 5, 6, 7] while researches on multi-modalities across images, videos, audios and texts are insufficient. For fine-grained image retrieval, it aims to retrieve images belonging to multiple subordinate categories of a super-category (*aka* a meta-category), *e.g.*, different species of animals or plants [8]. The key challenge is therefore to understand the fine-grained visual differences in order to adequately distinguish different objects that are highly similar in overall appearance but differ in fine-grained patterns [9, 10, 11]. Thus, integrating fine-grained data from four mentioned modalities for a unified retrieval will produce great value in various applications. However, the issues caused by fine-grained nature in real-life across multi-modalities cannot be addressed by traditional coarse-grained cross-modal retrieval methods directly. Nowadays, with the explosion of fine-grained data in real-world applications [9], fine-grained sub-category retrieval across multi-modalities has become an urgent technical requirement in cross-modal retrieval.

Generally, fine-grained cross-modal retrieval faces two main challenges: (1) Small inter-class variations but large inter-modal variations - Each modality has highly similar subcategories while data representation among different modalities varies considerably. (2) Large intra-class variations but small intra-modal variations - Each sub-categories has large intra-class variations in poses, scales and rotations while data representation within the modalities has small differences.

In response to the aforementioned issues, this paper focuses on the following aspects: In order to alleviate the problem of inter-modal variability, a channel mix method is proposed, which can enhance information interaction in different modalities for fine-grained objects and reduce the variations among different modalities. Then, a loss function for fine-grained cross-modal retrieval is developed to further improve the intra-class separability as well as inter-class compactness for multi-modalities. Experiments are conducted on the fine-grained cross-modal benchmark dataset, *i.e.*, FG-XMedia [12]. Empirical results and ablation studies could demonstrate the effectiveness of our method, as well as its proposals.

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2. RELATED WORK

2.1. Fine-Grained Retrieval

Fine-grained retrieval is a fundamental topic of fine-grained image analysis [9, 10, 13, 14, 15]. Compared with generic image retrieval, which focuses on retrieving near-duplicate images based on similarities in their content (*e.g.*, texture, color, and shape), fine-grained retrieval aims to retrieve the images of the same category type (*e.g.*, the same subordinate species of animal). Depending on the types of query images, fine-grained image retrieval tasks can be separated into two groups, *i.e.*, fine-grained content based image retrieval (FG-CBIR) and fine-grained sketch-based image retrieval (FG-SBIR). In concretely, SCDA [16] is one of the earliest work of FG-CBIR that used deep learning without using explicit localization supervisions. Supervised metric learning based approaches were then proposed to overcome the retrieval accuracy limitations of unsupervised retrieval [10, 11]. FG-SBIR is another interesting task related to both fine-grained image retrieval and cross-modal retrieval of which goal is to match specific photo instances using a free-hand sketch as the query modality. Existing FG-SBIR approaches generally aim to train embedding space where sketches and photos can be compared in a nearest neighbor fashion [15, 13]. While, the requirements in fine-grained retrieval techniques exist not only for the image and sketch modalities, but also for other modalities such as videos and texts.

2.2. Cross-Modal Retrieval

Cross-modal retrieval also leads to a birth of exciting tasks in recent years [2, 17, 18, 3, 19]. The main approach of cross-modal retrieval is to map features from different modalities into a common space for representation learning [4, 2]. There were also some methods using generative adversarial networks to achieve progress [17]. Hashing based cross-modal retrieval [18] is another meaningful sub-topic which dedicates to helping save storage space and reducing retrieval cost. However, existing researches generally focus on coarse-grained cross-modal retrieval while few focuses on fine-grained cross-modal retrieval. Specifically, [3] established strongly supervised learning between images and texts in different parts by performing multi-modal representation learning and used it for fine-grained cross-modal retrieval. For the datasets for fine-grained cross-modal retrieval, [12] constructed a benchmark which contains four modalities, *i.e.*, texts, images, audios and videos, based on the fine-grained image dataset *CUB200-2011* [20] and the fine-grained video dataset *YouTube Birds* [21]. Furthermore, in [12], the authors also proposed an approach for handling four modalities simultaneously.

2.3. Mixup Methods

Mixup methods [22, 23, 24], usually known as synthesizing training data or data augmentation strategies, are popular used in image recognition, image retrieval and object detection tasks. Mixup [22] and its varients [23] aim at performing pixel-level or feature-level interpolation to benefit deep neural networks learn better representations. In details, for the lack of a deep analysis in particular on the localization ability and transfer learning performance, CutMix [25] was purposed to overcome these problems. Then, FMix [26] argued that an mixed sample data augmentation (MSDA) which builds on CutMix [25] to include masks of arbitrary shapes, rather than just squares, could further prevent memorisation whilst preserving the data distribution in the same way. However, all these mixup methods were performed in the pixel level or feature level space for mixup training data. In this paper, we attempt to discover that whether a channel mix like method can improve the retrieval accuracy based on exchanging information from the channels of deep activations across multi-modalities.

3. APPROACH

3.1. Data Preprocessing and Backbone Networks

Data Preprocessing In order to take data from different modalities as the inputs of the vanilla backbone, we need to conduct data preprocessing firstly. For the image modality, there is no preprocessing. For the audio modality, we apply Short-Time Fourier Transformation to generate spectrograms for each audio instance referring to [27]. For the text modality, we use the pre-trained model of GloVe [28] to extract a 250×300 dimensional sentence vector for each text instance. For the video modality, we extract k key frames which is described in Section 4.2.

Backbone Networks The main framework of our proposed method is shown in Figure 1. For the video, image and audio modalities, we use ResNet-50 [29] separately as the backbone model. For the text modality, we construct a model with only two layers of convolutional networks by following TextCNN [30]. It is noted that these adopted backbones can be replaced by any other state-of-the-art deep convolutional neural networks.

3.2. Cross-Modal Channel Mix

The representation learning component plays an important role in cross-modal retrieval [4, 2]. High-order feature interactions contribute to generate discriminative feature representations between fine-grained categories [11]. However, deep features from different modalities do not have any interactions before feeding into the representation learning component. Thus, we propose Cross-Modal Channel Mix (CMCM) to perform channel interactions upon representation learning across different modalities and generate discriminative deep features between fine-grained sub-categories. The main goal of CMCM is to make interactions among all modalities which will enable the deep features of each modality to carry rich semantic information from different modalities during the training process.

In the four given modalities $\mathcal{M} = \{\mathcal{I}, \mathcal{A}, \mathcal{V}, \mathcal{T}\}$, assuming the category space to be $\mathcal{Y} = \{y_1, y_2, \ldots, y_N\}$ and the sam-

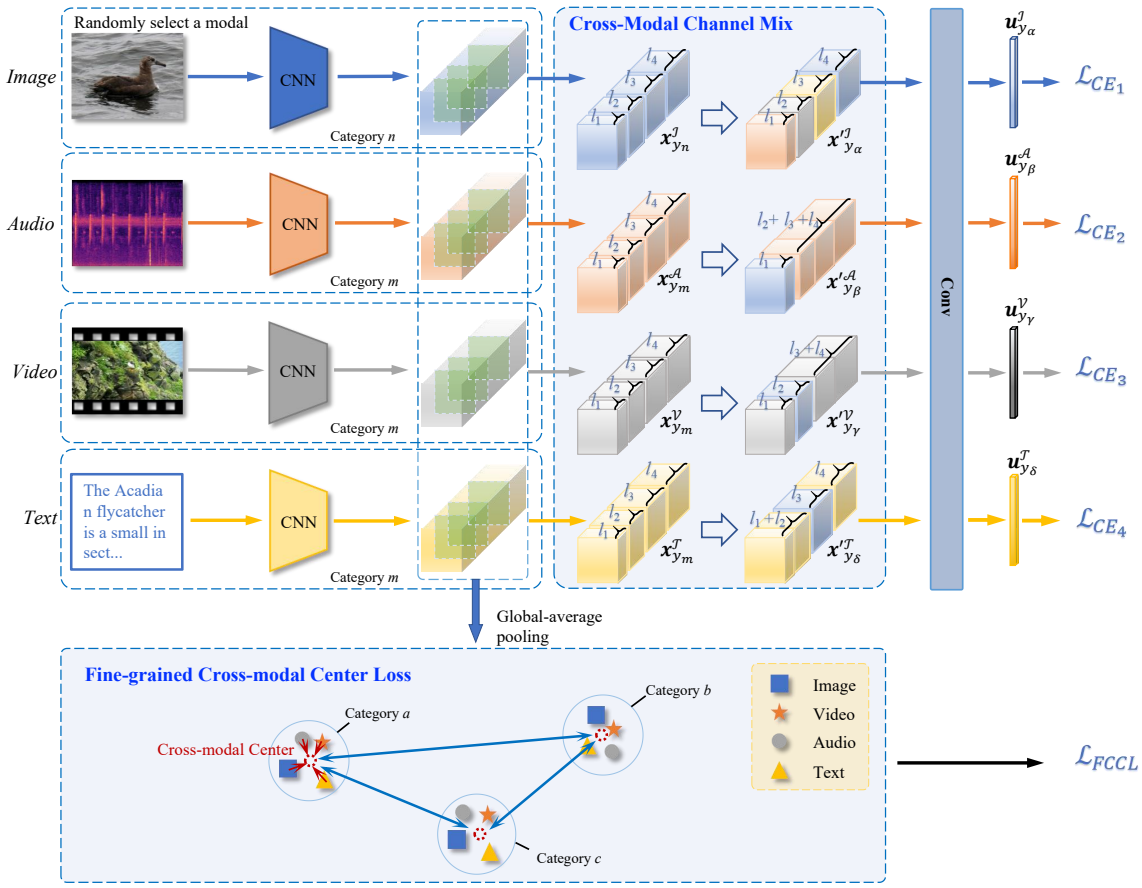

Fig. 1. Overview framework of our proposed Cross-Modal Channel Mix (CMCM) method.

ple space to be $\Omega = \{\mathcal{X}^{\mathcal{M}}_{y_1}, \mathcal{X}^{\mathcal{M}}_{y_2}, \ldots, \mathcal{X}^{\mathcal{M}}_{y_N}\}$, where $\mathcal{X}^{\mathcal{M}}_{y_i} = \{\mathcal{X}^{\mathcal{I}}_{y_i}, \mathcal{X}^{\mathcal{A}}_{y_i}, \mathcal{X}^{\mathcal{V}}_{y_i}, \mathcal{X}^{\mathcal{T}}_{y_i}\}$ is a collection of instances in modality $\mathcal{M}$ belongs to category y_i. During the training data selection stage, we randomly select a category y_m. For each modality in $\mathcal{M}$, we then randomly select a sample in $\mathcal{X}^{\mathcal{M}}_{y_m}$ to form an image-audio-video-text set $P^{\mathcal{M}}_{y_m} = \{P^{\mathcal{I}}_{y_m}, P^{\mathcal{A}}_{y_m}, P^{\mathcal{V}}_{y_m}, P^{\mathcal{T}}_{y_m}\}$, where $P^{\mathcal{I}}_{y_m} \in \mathcal{X}^{\mathcal{I}}_{y_m}$ is an image sample belonging to category y_m and the same with other modalities. We extract the deep features set $\boldsymbol{x}^{\mathcal{M}}_{y_m} = \{\boldsymbol{x}^{\mathcal{I}}_{y_m}, \boldsymbol{x}^{\mathcal{A}}_{y_m}, \boldsymbol{x}^{\mathcal{V}}_{y_m}, \boldsymbol{x}^{\mathcal{T}}_{y_m}\}$ of the input image-audio-video-text sample $P^{\mathcal{M}}_{y_m}$ via a corresponding backbone CNN model $\Phi^{\mathcal{M}}_{\text{CNN}}(\cdot)$ by

$$\boldsymbol{x}^{\mathcal{M}}_{y_m} = \Phi^{\mathcal{M}}_{\text{CNN}}(P^{\mathcal{M}}_{y_m}) \in \mathbb{R}^{d \times H \times W}, \tag{1}$$

where d is the feature dimensions set with, in this work, $d^{\mathcal{I}} = d^{\mathcal{A}} = d^{\mathcal{V}} = d^{\mathcal{T}} = d$.

To perform CMCM in the training stage, we randomly select a modality in $\mathcal{M}$ and replace the related deep feature with another sample in the selected modality belongs to different category in the image-audio-video-text deep feature set $\boldsymbol{x}^{\mathcal{M}}_{y_m}$. Taking the replacement of image modality as an example, the deep feature $\boldsymbol{x}^{\mathcal{I}}_{y_m} \in \boldsymbol{x}^{\mathcal{M}}_{y_m}$ will be replaced by a deep feature extracted from another sample in $\mathcal{X}^{\mathcal{I}}_{y_n}$ belonging to category y_n in the same training mini-batch. Thus, the new image-audio-video-text set of deep features is donated as $\boldsymbol{x} = \{\boldsymbol{x}^{\mathcal{I}}_{y_n}, \boldsymbol{x}^{\mathcal{A}}_{y_m}, \boldsymbol{x}^{\mathcal{V}}_{y_m}, \boldsymbol{x}^{\mathcal{T}}_{y_m}\}, y_m \neq y_n$. Then, we split each deep feature in the new set $\boldsymbol{x}$ into four parts:

$$\boldsymbol{x}^{\mathcal{I}}_{y_n} = [(\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_1)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_2)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_3)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_4)}], \tag{2}$$

$$\begin{aligned}\boldsymbol{x}^{\{\mathcal{A},\mathcal{V},\mathcal{T}\}}_{y_m} = [&(\boldsymbol{x}^{\{\mathcal{A},\mathcal{V},\mathcal{T}\}}_{y_m})_{(l_1)}; (\boldsymbol{x}^{\{\mathcal{A},\mathcal{V},\mathcal{T}\}}_{y_m})_{(l_2)}; \\ &(\boldsymbol{x}^{\{\mathcal{A},\mathcal{V},\mathcal{T}\}}_{y_m})_{(l_3)}; (\boldsymbol{x}^{\{\mathcal{A},\mathcal{V},\mathcal{T}\}}_{y_m})_{(l_4)}],\end{aligned} \tag{3}$$

where l_1, l_2, l_3, l_4 are the length of each part and their values are randomly determined in each mini-batch with $l_1 + l_2 + l_3 + l_4 = d$. Recombining these deep features in the image-audio-video-text set $\boldsymbol{x}$ could obtain new sample representations:

$$\boldsymbol{x}'^{\mathcal{I}}_{y_\alpha} = [(\boldsymbol{x}^{\mathcal{A}}_{y_m})_{(l_1)}; (\boldsymbol{x}^{\mathcal{V}}_{y_m})_{(l_2)}; (\boldsymbol{x}^{\mathcal{T}}_{y_m})_{(l_3)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_4)}], \tag{4}$$

$$\boldsymbol{x}'^{\mathcal{A}}_{y_\beta} = [(\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_1)}; (\boldsymbol{x}^{\mathcal{A}}_{y_m})_{(l_2)}; (\boldsymbol{x}^{\mathcal{A}}_{y_m})_{(l_3)}; (\boldsymbol{x}^{\mathcal{A}}_{y_m})_{(l_4)}], \tag{5}$$

$$\boldsymbol{x}'^{\mathcal{V}}_{y_\gamma} = [(\boldsymbol{x}^{\mathcal{V}}_{y_m})_{(l_1)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_2)}; (\boldsymbol{x}^{\mathcal{V}}_{y_m})_{(l_3)}; (\boldsymbol{x}^{\mathcal{V}}_{y_m})_{(l_4)}], \tag{6}$$

$$\boldsymbol{x}'^{\mathcal{T}}_{y_\delta} = [(\boldsymbol{x}^{\mathcal{T}}_{y_m})_{(l_1)}; (\boldsymbol{x}^{\mathcal{T}}_{y_m})_{(l_2)}; (\boldsymbol{x}^{\mathcal{I}}_{y_n})_{(l_3)}; (\boldsymbol{x}^{\mathcal{T}}_{y_m})_{(l_4)}]. \tag{7}$$

Then, a 1×1 convolution layer is used to aggregate channel information and an average pooling layer is used to

extract the final feature vectors $\boldsymbol{u} = (\boldsymbol{u}^{\mathcal{I}}_{y_\alpha}, \boldsymbol{u}^{\mathcal{A}}_{y_\beta}, \boldsymbol{u}^{\mathcal{V}}_{y_\gamma}, \boldsymbol{u}^{\mathcal{T}}_{y_\delta})$ from the new image-audio-video-text deep features set $\boldsymbol{x}' = \{\boldsymbol{x}'^{\mathcal{I}}_{y_\alpha}, \boldsymbol{x}'^{\mathcal{A}}_{y_\beta}, \boldsymbol{x}'^{\mathcal{V}}_{y_\gamma}, \boldsymbol{x}'^{\mathcal{T}}_{y_\delta}\}$. Referring to [25], the corresponding soft labels of these vectors are defined as below:

$$y_\alpha = \frac{d - l_4}{d} y_m + \frac{l_4}{d} y_n \,, \tag{8}$$

$$y_\beta = \frac{d - l_1}{d} y_m + \frac{l_1}{d} y_n \,, \tag{9}$$

$$y_\gamma = \frac{d - l_2}{d} y_m + \frac{l_2}{d} y_n \,, \tag{10}$$

$$y_\delta = \frac{d - l_3}{d} y_m + \frac{l_3}{d} y_n \,. \tag{11}$$

The process of CMCM maintains positional information of deep features thus the original semantic information will not be disrupted. In the training stage, we use the cross entropy loss to constrain the final feature vectors of each modality. Since the representation of each modality is reassembled and masked with a new soft label, its cross entropy loss also follows the relevant definition in CutMix [25].

3.3. Fine-Grained Cross-Modal Center Loss

To obtain a better performance in fine-grained cross-modal retrieval, in addition to constraining the feature vectors with the same category in each modality, it is also important to pay attention to the spatial distance of feature vectors which belongs to different modalities with the same category. On the other hand, when minimizing the spatial distance among different modalities in the same category, it is also necessary to keep the spatial distance among different modalities in different categories. Thus, the proposed Fine-grained Cross-modal Center Loss (FCCL) can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{FCCL} =& \frac{\vartheta}{2} \sum_{i=1}^{b} \|\boldsymbol{u}_i^{\cdot} - \boldsymbol{c}_i\|_2^2 + \\ & \frac{\zeta}{2} \sum_{i=1}^{b} \sum_{j=1}^{b} \left[\xi - \mathbb{1}_{\Psi(\boldsymbol{u}_i^{\cdot}, \boldsymbol{u}_j^{\cdot}) > \sigma} \cdot \|\boldsymbol{c}_i - \boldsymbol{c}_j\|_2^2 \right]_+ ,\end{aligned} \tag{12}$$

where b is the size of mini-batch, $\boldsymbol{u}_i^{\cdot}$ represents the ith feature vector in each modality among the training data within the mini-batch. $\boldsymbol{c}_i$ is the inner-modality center related to $\boldsymbol{u}_i^{\cdot}$. $\Psi(\cdot)$ represents the cosine similarity function, $\mathbb{1}(\cdot)$ represents the indicator function while $[\cdot]_+$ indicates the hinge function $\max(0, \cdot)$. ξ is a margin that is enforced between each two categories. ϑ, ζ and σ are hyper-parameters.

In the training stage, we use both cross entropy loss and fine-grained cross-modal center loss to drive the network training. The final loss being minimized is presented as

$$\mathcal{L} = \mathcal{L}_{FCCL} + \sum_{i=1}^{4} \mathcal{L}_{CE_i} \,. \tag{13}$$

4. EXPERIMENTS

4.1. Datasets

The dataset used in experiments is currently the only available dataset for fine-grained cross-modal retrieval, *i.e.*, the FG-Xmedia dataset [12]. The dataset contains samples from four modalities of 200 fine-grained sub-categories, including 11,788 images, 8,000 texts, 18,350 videos, and 12,000 audios. We divide the dataset by following [12] for fair comparison. For the image modality, the FG-Xmedia dataset directly use images from *CUB200-2011* [20], therefore the training set contains 5,994 images while the test set contains 5,794 images. For the audio modality, both the training set and the test set contain 6,000 audios. For the text modality, both the training set and the test set contain 4,000 texts. For the video modality, the FG-Xmedia dataset use videos collected in *YouTube Birds* [21]. For there are 1,214 videos not available, we use the rest 11,813 videos as the training set, and 5,323 videos as the test set.

4.2. Baselines and Implementation Details

Baselines In experiments, we compare our proposed method to the following competitive baselines, *i.e.*, GXN [1], SCAN [2], CMDN [4], GSPH [18], JRL [5], ACMR [6], MHTN [7], and FGCrossNet [12]. GXN [1] and SCAN [2] are specially designed for cross-modal retrieval between image and text modalities. CMDN [4] learns separate representations for each modality by multiple deep networks. GSPH [18] proposes a hashing method to preserve the semantic distance between each two modality. JRL [5] applies semi-supervised regularization in cross-modal retrieval while ACMR [6] uses adversarial learning method. MHTN [7] transfers knowledge form single-modal source domain (image) to cross-modal target domain. FGCrossNet [12] simultaneously learns four types of modality without discriminative treatments.

Implementation Details In this part, we will firstly describe the k key frames in each video. Taking action-based recognition for example, we cannot tell whether the action is a wave or a clap by only one frame in the video. It is necessary to play the complete or majority of the whole action before we make final decision. However, fine-grained cross-modal retrieval for a typical object is of great difference. For example, we cannot decide which category the bird is belonged to by noticing them eating or walking on the lawn. Therefore, we need to extract as many key frames as possible which contain both local and global attributes of each category as the overall description of the given videos. Thus, in the training stage, we divide each video into 25 frames equally. Then a pre-trained network, in this work we use ResNet-50 [29], is used to extract feature vectors of each frame. The parameters of this network are frozen during the feature extraction process so that it will not cost much time. For each given video, a center feature vector is calculated according to these extracted vectors and φ key frames of which feature vectors are nearest to the center feature

Table 1. Comparisons of bi-modality fine-grained cross-modal retrieval accuracy (% mAP) on all four modalities within the FG-Xmedia dataset, including image, text, video and audio.

Methods	I→T	I→A	I→V	T→I	T→A	T→V	A→I	A→T	A→V	V→I	V→T	V→A	Average
CXN	0.023	-	-	0.035	-	-	-	-	-	-	-	-	0.029
SCAN	0.050	-	-	0.050	-	-	-	-	-	-	-	-	0.050
CMDN	0.099	0.009	0.377	0.123	0.007	0.078	0.017	0.008	0.010	0.446	0.081	0.009	0.105
GSPH	0.140	0.098	0.413	0.179	0.024	0.109	0.129	0.024	0.073	0.512	0.126	0.086	0.159
JRL	0.160	0.085	0.435	0.190	0.028	0.095	0.115	0.035	0.065	0.517	0.126	0.068	0.160
ACMR	0.162	0.119	0.477	0.075	0.015	0.081	0.128	0.028	0.068	0.536	0.138	0.111	0.162
MHTN	0.116	0.195	0.281	0.124	0.138	0.185	0.196	0.127	0.290	0.306	0.186	0.306	0.204
FGCrossNet	0.210	0.526	**0.606**	0.255	0.181	0.208	0.553	0.159	0.443	**0.629**	0.195	0.437	0.366
Ours	**0.349**	**0.586**	0.597	**0.353**	**0.284**	**0.283**	**0.592**	**0.281**	**0.450**	0.625	**0.323**	**0.486**	**0.434**

Table 2. Comparisons of multi-modality fine-grained cross-modal retrieval accuracy (% mAP) on the FG-Xmedia dataset.

Method	I→All	T→All	V→All	A→All	Average
ACMR	0.245	0.039	0.041	0.279	0.151
GMDN	0.321	0.071	0.016	0.229	0.159
JRL	0.344	0.08	0.069	0.275	0.192
CSPH	0.387	0.103	0.075	0.312	0.219
MHTN	0.208	0.142	0.237	0.341	0.232
FGCrossNet	0.549	0.196	0.416	0.485	0.412
Ours	**0.599**	**0.291**	**0.516**	**0.496**	**0.476**

vector are selected to participate in the update of parameters in the network.

In the training stage, the probability of adopting CMCM is 0.5, which means each mini-batch have a 50% chance of training with channel mix. ϑ, ζ is set as 0.01, σ is set as the average similar size of each sample among all modalities within the same mini-batch while ξ is the average Euclidean distance between each sample within the same mini-batch. φ is set as 5, which means for each video, we will retain 5 key frames. The total number of training epoch is 50 and the learning rate is set as 10^{-4}. The optimizer is standard mini-batch stochastic gradient descent with the weight decay as 10^{-4}. In the retrieval stage, the feature vector of each video is designed to be the arithmetic mean of feature vectors extracted from 5 key frames.

Evaluation Metric For multi-modal fine-grained retrieval tasks, the experiments apply the mean average precision (mAP) score to measure its performance. The mAP score is calculated by the mean value of the average precision (AP) of each query. To get a comprehensively understanding of the retrieval performance at all modalities, we perform one-to-one and one-to-all cross-modal retrieval experiments for each modality separately.

4.3. Main Results

Figure 2 gives an example of the sub-task I→All. Table 1 presents the mAP results of bi-modality fine-grained cross-modal retrieval on the FG-Xmedia dataset. As shown in that table, our proposed method significantly outperforms the other baseline methods on the FG-Xmedia dataset in many sub-tasks. In many scenarios such as I→T, T→I, T→A, A→T, T→V, V→T, the proposed method obtains 10% mAP improvement in average than previous state-of-the-art. Table 2 shows the mAP score of the one-to-all retrieval experiment in which a certain modality sample as a query to retrieve all modality samples. In this experiment, T→All has a big improvement of more than 45% compared with previous state-of-the-art while other scenarios have also improved significantly. The results above prove the effectiveness of our proposed approach at multi-modal fine-grained retrieval.

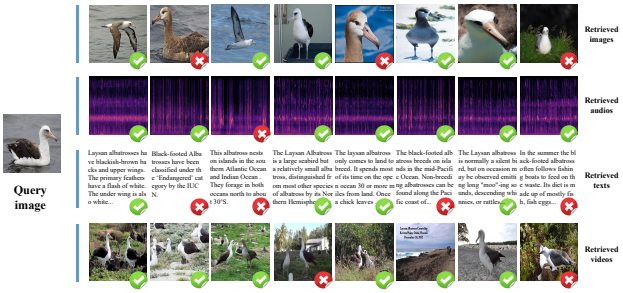

Fig. 2. Examples of top-8 retrieved samples from four modalities on FG-Xmedia by our method.

4.4. Ablation Studies

In this section, we demonstrate the effectiveness of these crucial components of the proposed proposals of CMCM, *i.e.*, the cross-modal channel mix component (cf. 3.2) and the fine-grained cross-modal center loss component (cf. Section 3.3). In the ablation studies, we apply these components incrementally on a vanilla backbone (cf. Section 3.1) as the baseline. As evaluated in Table 3, by stacking these two components one by one, the retrieval results are steadily improved in most of the modalities, which justifies the effectiveness of our proposed method.

Table 3. Multi-modality fine-grained cross-modal retrieval accuracy (% mAP) with incremental components of the proposed method.

Configurations	I→All	T→All	V→All	A→All	Average
Vanilla backbone	0.557	0.259	0.471	0.279	0.392
+ Channel mix	0.586	0.285	**0.519**	0.481	0.468
+ FCCL loss	**0.599**	**0.291**	0.516	**0.496**	**0.476**

5. CONCLUSION

In this paper, we proposed a channel mix method for handling fine-grained cross-modal retrieval. Specifically, for the channels of deep activations of different modalities, given one specific modality, we exchanged part of channels with the channels from other modalities, which could enhance information interaction in multi-modal data. Moreover, a fine-grained center-loss was designed to further improve both the intra-class separability and inter-class compactness of learned visual features. Experiments on the FG-Xmedia benchmark dataset validated the effectiveness of the proposed method. In the future, developing discriminative part-aware channel mix is promising for further boosting the performance.

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